Athena Stein

PhD MPH BBiomedSc FHEA

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LinkedIn Twitter

Research Profile

I am a postdoctoral researcher in psychiatry and translational neuroscience building an independent program in biomimetic neuromodulation. My research focuses on designing brain-derived stimulation waveforms that preserve endogenous temporal structure and validating target engagement with electrophysiology (i.e., high density EEG) to guide rational optimization of novel stimulation treatments. I have recently developed transcranial endogenous current stimulation (tECS), physiology-informed TMS (piTMS), and my recent work has involved conducting clinical trials using closed-loop tACS in major depressive disorder and examining intracranial electrophysiology in mood circuits.

Education and Training

Doctor of Philosophy (PhD), Clinical Neuroscience **Oct 2020 – Apr 2024 (Conferred)**
The University of Queensland Child Health Research Centre, South Brisbane.
Thesis: *"Investigating the use of non-invasive brain stimulation to improve attention in paediatric acquired brain injury"*
Advisor: Prof. Karen Barlow *PhD, MSc, MBChB, MRCPCH, FRACP*

- Conducted mechanistic investigations HD-EEG, fMRI, and DTI/NODDI, as well as two clinical trials to study mechanisms of recovery and response to non-invasive brain stimulation in pediatric brain injury.
- Examined structural and functional biomarkers associated with interindividual variability in tDCS response.
- Developed and implemented a home-based, remotely supervised tDCS clinical trial in pediatric brain injury.

Master of Public Health (Global Health) **Jan 2019 – Jul 2020**

The University of Queensland, Herston
Thesis: "*Access to Deep Brain Stimulation Treatment for Patients with Parkinson's Disease: A Comparative Policy Analysis of Fourteen High-Income Countries*" (see publications)

- HRH Duchess of Gloucester Prize in Global Health (2020)

Bachelor of Biomedical Science, Neuroscience **Jan 2016 – Dec 2018**
The University of Queensland, St Lucia;

- Queensland Brain Institute Research Intern, high distinction (2018)
- School of Chemistry Undergraduate Research Program, high distinction (2016)

Academic Appointments

Postdoctoral **Jul 2024 – present**
University of North Carolina at Chapel Hill (Reference: Flavio Frohlich, flavio_frohlich@med.unc.edu)

- Developed biomimetic neuromodulation approaches in psychiatry, including transcranial endogenous current stimulation (tECS), physiology-informed TMS (piTMS)
- Leading scientific development and implementation of clinical trials across three translational neuromodulation studies, including closed-loop tACS in major depressive disorder and intracranial electrophysiology projects.
- Conducting high-density EEG and source-localized connectivity analyses to define target engagement mechanisms.
- Primary mentor/advisor for 9 undergraduate research assistants and 3 full-time study coordinators (clinical trial team)

Honorary Research Fellow **Aug 2024 – present**
University of Queensland Faculty of Medicine (Reference: Craig Munns, c.munns@uq.edu.au)
Advising on translational non-invasive brain stimulation projects in pediatric acquired brain injury.

Senior Research Assistant **Jul. 2020 – Jul 2024**
University of Queensland Child Health Research Centre (Reference: Karen Barlow, k.barlow@uq.edu.au)

- rTMS and EEG technician in pediatric neuromodulation and neurophysiology research across concussion, autism, and acquired brain injury studies

Senior Research Assistant **Aug 2022 – Dec 2022**
UQ School of Public Health (Reference: Marina Reeves, m.reeves@sph.uq.edu.au)

- Mixed methods analysis of longitudinal data on student feedback following redesign of PUBH7000 (Research Methods in MPH degree)

Contracted Academic Editor **Jul. 2019 – Oct 2020**
Enago (Reference: Veronica Jacob, veronicaj@enago.com)
Uni-edit (Reference: Daniel Boles, daniel.uniedit@gmail.com)

- **Specialisations:** biomedical science; biology; neuroscience; public health; social science; humanities; policy
- Edited over 70 manuscripts for publishing in international journals including BMJ open; PLOS ONE; AJRCMB

Summer Research Scholarship recipient **Nov. 2019 – Feb. 2020**
UQ Child Health Research Centre (Reference: Kartik Iyer, k.iyer@uq.edu.au)

- Led a task fMRI analysis of working memory in 90 children with acquired brain injury and healthy controls

Teaching Experience

Teaching Assistant, PUBH7000 (Research Methods for Master of Public Health) **Feb. – Jun. 2021- 2023**
The University of Queensland (Reference: Marina Reeves, m.reeves@sph.uq.edu.au)

- Facilitating 3 hour weekly workshops
- Delivered guest lectures and workshops each semester
- *Received a score of **5/5 for SECAT student teaching evaluations each year 2021- 2023***
- *Appointed **Fellow of the Higher Education Academy (FHEA), 2021**: Demonstrate substantial, evidence-based teaching experience in higher education, aligned with the Professional Standards Framework*

Co-Course Coordinator, BIOM1061 (Pharmacology for Nursing and Midwifery **Nov. 2020 – Jan 2021**
The University of Queensland (Reference: Mary-Louise Manchadi, m.roymanchadi@uq.edu.au)

- Co-coordinated summer course

Tutor and course development consultant, BIOM3200 (Biomed. Sci. Capstone Course) **Aug. – Oct. 2020; 2021**
The University of Queensland (Reference: Jo Bowles, jo.bowles@uq.edu.au)

- Played a lead role in re-development of Bachelor of Biomedical Science Research Capstone course

Research Funding (USD ~$435,000)

Date	Funding Source	Role / Purpose	USD	Original (AUD)
Research Grants				
Apr 2026	Foundation of Hope: Impact Grant	PI, piTMS clinical trial	$200,000	
Jan 2025	Brain Behavior Research Foundation (BBRF) NARSAD Distinguished Investigator Grant	Co-I; writing lead; tECS clinical trial	$100,000	
Nov 2021	Small Equipment Grant, UQ Child Health Research Centre	Primary applicant; funded fMRI-compatible tDCS/tACS device	$6,600	$10,000
Sep 2021	Small Research Grant, UQ Child Health Research Centre	Primary applicant; study costs	$3,300	$5,000
Aug 2021	Pilot Scheme, Herston Imaging Research Facility (HIRF)	Co-I; awarded 10 MRI scans	$4,290	$6,500
Fellowships and Scholarships				
Jul 2024	American Australian Association Graduate Education Scholarship	Primary sole applicant	$40,000	
Oct 2020 - Apr 2024	Australian Government Research Training Program (RTP) Scholarship	Living stipend (approx. $32,000 AUD per year for 3.5 years)	$73,920	$112,000
Travel Awards				
Jan 2026	Society for Biological Psychiatry	Travel Award to SOBP 2026 NYC	$2,000	
Sep 2023	UQ Faculty of Medicine Thomas McIntyre Scholarship	Travel to ABSS and ACNS (Sydney, Australia)	$1,310	$1,990
Feb 2023	UQ Faculty of Medicine Researcher Development Award	Travel to International Brain Stimulation Conference (Portugal)	$2,640	$4,000
Aug 2022	Abbott Science and Society Award for Emerging Scientists	Travel to NYC Neuromodulation	$1,000	

Honors and Awards

Academic Awards and Fellowships

- Society for Biological Psychiatry (SOBP) Travel Award Recipient (2026)
- Runner Up, PhD Graduate of the Year, UQ Faculty of Medicine (2024)
- Student of the Year, UQ Child Health Research Centre (2023)
- HRH Princess Alice, Duchess of Gloucester Prize in Global Health – Highest grade in Global Health (2020)
- Dean's Academic Commendation, UQ School of Public Health (2020)
- UQ Summer Research Scholarship – Child Health Research Centre (2019–2020)

Public Engagement and Teaching Recognition

- Fellow of the Higher Education Academy (FHEA) – UK Professional Standards Framework (2021)
- Teaching and Learning Excellence Award, UQ Faculty of Medicine (2021)

Student Competitions and Popular Vote Awards

- People's Choice Presentation, UniQuest Commercialisation Workshop (2023)
- People's Choice Presentation, UQ Three Minute Thesis School/Institute Final (2023)
- Highly Commended Presentation, UQ CHRC HDR/EMCR Showcase (2023)
- Best Runner Up, UQ Faculty of Medicine HDR Symposium (2021)

Selected Invited Talks and Presentations

Invited Talks and Seminars

1. Invited Speaker — American Australian Association Annual Benefit Dinner, New York City, USA (Oct 2024)
2. Invited Speaker — *UC & Western Neurotrauma Symposium*, UCLA, Los Angeles, USA (Oct 2024). *"Developing tDCS as a treatment for attention problems in pediatric acquired brain injury: From biomarker identification to at-home treatment"*
3. Invited Seminar — *UC San Diego (USA) and CAMH Toronto (Canada)* (Jul 2022). *"Using non-invasive brain stimulation to improve executive function following acquired brain injury in children"*

Symposia and Oral Presentations

4. Symposium Chair, Organizer, and Speaker — *6th International Brain Stimulation Conference*, Kobe, Japan (Feb 2025). *"At-home brain stimulation in complex clinical populations: Practical, clinical, and scientific considerations"*
5. Symposium Co-Chair and Organizer — *6th International Brain Stimulation Conference*, Kobe, Japan (Feb 2025). *"Closed-Loop Brain Stimulation – Pushing the Boundaries towards Clinical Applications through Multiple Modalities"*
6. Oral Presentation — *Australasian Brain Stimulation Society (ABSS)*, Sydney, Australia (Nov 2023). *"Using tDCS to improve attention in children with acquired brain injury: Correlates of response and home-based treatment"*

Oral and Poster Presentations

7. Oral Presentation – Neuromodec NYC Neuromodulation (Aug 2026): *"Establishing Transcranial Endogenous Current Stimulation (tECS): A Pilot Study Modulating the EEG Aperiodic Exponent with Biomimetic Stimulation"*
8. Poster Presentation – SOBP Conference, NYC, USA (Apr 2026). *"The Time Course of Neural Target Engagement in a Randomized Trial of 10 Hz Tacs for Depression: Daily EEG Alpha Network Dynamics across Five Days"*
9. Poster Presentation – SOBP Conference, Toronto, Canada (Apr 2025). *"Daily changes in high-density EEG alpha power and functional connectivity across five consecutive days of 10Hz transcranial alternating current stimulation in depression: Results from a randomized controlled trial"*
10. Poster Presentation — *5th International Brain Stimulation Conference*, Lisbon, Portugal (Feb 2023). *"tDCS at home in children with acquired brain injury (hrtDCS-Attention): A dose-controlled clinical trial protocol"*

Publication Record (12 first author, 1 senior author)

Manuscripts

1. **Stein A**, Floegel E, Zhang M, Subramaniyan A, Sharma S, Feldman Z, Evers M, Riddle J, Meltzer-Brody S, Frohlich F. (2026). *"Establishing naturalistic brain stimulation targeting aperiodic EEG features: Transcranial Endogenous Current Stimulation (tECS)"*. Brain Stimulation: Basic, Translational, and Clinical Research in Neuromodulation. https://doi.org/*10.1016/j.brs.2026.103036*
   - *Role: Conceptualization, developed novel stimulation technique, analyzed all data (high-density EEG power, connectivity analyses), supervised data collection (primary mentor for trial team), wrote original draft*

2. Friehs, M.A., Schroeder, P.A., Barlow, K.M., **Stein, A.** (2026) *"Not just childish games: An exploration of different game-based assessments and discussion of gamification in clinical pediatric populations".* J Cogn Enhanc. https://doi.org/10.1007/s41465-026-00345-9
   - *Role: Conceptualization, collected all data, analyzed all data (task performance, clinical outcomes), wrote 50% of original draft*

3. **Stein A**, Riddle J, Caulfield KA, Dux PE, Friehs MA, Schroeder PA, Craven MP, Groom MJ, Iyer KK, Barlow KM (2025). *"Safety, Feasibility, and Tolerability of Ten Days of At-Home, Remotely Supervised tDCS During Gamified Attention Training in Children with Acquired Brain Injury: An Open-Label, Dose-Controlled Pilot Trial."* Brain Sciences. 2025 May 24;15(6):561. https://doi.org/10.3390/brainsci15060561
   - *Role: Conceptualization, collected all data, analyzed all data (HD-EEG source localized connectivity, clinical outcomes), wrote original draft*

4. **Stein A,** Caulfield KA, Singh M, Riddle J, Friehs MA, Craven MP, Groom MJ, Iyer KK, Barlow KM (2025). "*The Effect of a Single Session of Transcranial Direct Current Stimulation on Attention in Pediatric Acquired Brain Injury: Characterizing Interindividual Structural and Functional Network Response Variability.* Pediatric Neurology. *2025 Sep;170:133-45.* https://doi.org/10.1016/j.pediatrneurol.2025.06.005
   - *Role: Conceptualization, collected all data, analyzed all data (HD-EEG source localized connectivity, clinical outcomes), wrote original draft*

5. **Stein A***, Barlow KM* (2025). *"Orthostatic tachycardia in children with and without persisting post concussion symptoms following mild traumatic brain injury: A prospective controlled study".* Pediatric Neurology. *2025 Nov 7.* https://doi.org/10.1016/j.pediatrneurol.2025.11.001
   - *Role: Analyzed all data (physiological measurements, clinical outcomes), wrote original draft*

6. **Stein A,** Higgins N, Gajwani M, Gericke CA (2024). "*The treatment gap for deep brain stimulation in Parkinson's disease: a comparative analysis of cost and utilisation in high-income countries".* Australian Health Review. *2024 Sep 16;48(5):497-505.* *https://doi.org/10.1071/AH24036*
   - *Role: Collected all data (secondary data), analyzed data (cost, utilization, access data comparative analysis), wrote original draft*

7. **Stein, A**., Thorstensen, J.R., Ashley, D.P., Ho, J. M., Iyer, K. K., Barlow, K. M. (2024). *"Attention please! Unravelling the link between brain network connectivity and cognitive attention following acquired brain injury: a systematic review of structural and functional measures."* Brain Connectivity. https://doi.org/10.1089/brain.2023.0067
   - *Role: Conceptualization, led data collection (systematic review), analyzed all data (systematic review and quality analysis), wrote original draft*

8. **Stein, A.,** To, X.V., Nasrallah, F.A., Barlow, K.M. (2023). *"Evidence of ongoing cerebral microstructural reorganization in children with persisting symptoms following mild traumatic brain injury: a NODDI DTI analysis."* J Neurotrauma. https://doi.org/10.1089/neu.2023.0196
   - *Role: Analyzed all data (diffusion tensor imaging, predictive models, clinical outcomes), wrote original draft*

9. **Stein, A**., Iyer, K.K., Barlow, K.M. (2022). *"The effect of pediatric traumatic brain injury on simulated transcranial direct current stimulation electric field"* [conference paper]. Neuroergonomics and Neuromodulation Meeting Jul 2022: New York City, New York. https://shorturl.at/2fjIO
   - *Role: Analyzed all data (electric field modelling), wrote original draft*

10. **Stein, A.**, Iyer, K.K., Khetani, A., Barlow, K.M. (2021) *"Changes in Working Memory-Related Cortical Responses Following Pediatric Mild Traumatic Brain Injury: A Longitudinal Study"*. J Concussion, 5, p 1-15. *https://doi.org/10.1177/20597002211006541*
    - *Role: Analyzed all data (task functional MRI, predictive models, clinical outcomes), wrote original draft*

Book chapters

11. **Stein, A**., & Frohlich, F. (in press). "*Making waves: Overview and methodological considerations for the use of tACS in clinical and cognitive neuroscience applications".* In. E. G. Chrysikou (Ed.) Neuroscience Methods. New York: Springer Nature.
    - *Role: wrote 80% of original draft (review paper)*

12. **Stein, A**., Iyer, K. K., & Barlow, K. M. (2022). *"Cerebral activation of attention and working memory in traumatic brain injury."* In: Diagnosis and Treatment of Traumatic Brain Injury. Academic Press. p 151-167. https://doi.org/10.1016/B978-0-12-823347-4.00025-7
    - *Role: wrote original draft (review paper)*

Preprints

13. **Stein A***, Schwippel TU*, Pupillo FM, LaGarde HC, Zhang M, Rubinow DR, Frohlich F (2026). *"Daily EEG reveals stage-specific alpha power and functional connectivity modulation across five days of tACS in major depressive disorder."* medRxiv. 2026:2026-03. https://doi.org/10.64898/2026.03.17.26348546
    - *Role: Analyzed data (high density EEG power and connectivity analysis, behavioral outcomes), wrote original draft*

14. Schwippel T, Pupillo F, LaGarde H, **Stein A**, Zhang M, Rubinow D, Frohlich F (2026). *"Suppression of Endogenous Alpha Power Predicts Clinical Response to 10 Hz tACS in Major Depressive Disorder: A Double-Blind Randomized Controlled Trial."* medRxiv. 2026:2026-03. https://doi.org/10.64898/2026.03.17.26348625
    - *Role: Review and editing*

Mentoring and Supervision

Carolina Center for Neurostimulation, UNC-CH **Jul 2024 – present**

Lead mentor and supervisor for a clinical research team consisting of 9 undergraduate research assistants and 3 full-time study coordinators, across three brain stimulation clinical trials.

Co-advisor: PhD student, UQ Acquired Brain Injury in Children research group **Aug 2024 - present**

Efficacy of at-home tDCS to improve attention in children with acquired brain injury

Co-advisor: Honors student, UQ Acquired Brain Injury in Children research group **Jan 2024 -Dec 2024**

Analysis of brain connectivity in 6 year-old children born pre-term. Student outcome: First Class Honors

Service, Outreach, and Volunteering

Memberships

- Associate Member, Society for Research in Psychopathology **2025-present**
- Member, American Australian Association **2024 – present**
- Member, Graduate Women in Science (GWIS) **2024 – present**
- Member, CuttingEEG **2023 – present**
- Member, Australasian Brain Stimulation Society (ABSS) **2022 – present**
- Fellow, Higher Education Academy (FHEA) **2022 – present**

Professional Service

- UQ Rep, Biological Psychiatry Australia Committee **Jan. 2022 – Dec. 2022**
- Chair, UQ Faculty of Medicine HDR Student Committee **Jan. 2021 – Dec. 2021**
- Co-Chair, UQ CCHR HDR Student Committee **Jan. 2021 – Dec. 2021**

Science Communication

- Music and the brain (ongoing events – UNC) **Dec. 2024 – present**
- Science Ambassador – Wonder of Science QLD (UQ) **Mar. 2021 – 2024**
- Science Blog – A Stimulating Endeavour (Personal) **Sep. 2020 – Sep 2024**

Community Outreach

- **RSPCA** – Animal care assistant **Dec 2023 – Jul 2024**
- **Wesley Mission Asher House –** Care Volunteer, Acquired brain injury **Feb. 2019 – Nov. 2019**
  (Reference: Marie Duffy, MDuffy@wmq.org.au)
- **Queensland Children's Hospital –** Bedside Play Volunteer **Aug. 2017 – Feb. 2018**
  (Reference: Nicky Pearson, nicky.pearson@childrens.org.au)

Media engagement/Science Communication

The Rhythmic Forum, Carolina Center for Neurostimulation **April 2026**
I led coordination of, and performed in, this music and neuroscience mashup event on behalf of the Carolina Center for Neurostimulation featuring several musical performances, a neuroscience/mental health panel, and live EEG. Collaboration with Carolina Performing Arts and UNC School of Medicine.

Neuroscience and music event, Carolina Performing Arts/UNC School of Medicine **Feb 2025**
I led and coordinated the participation of Carolina Center for Neurostimulation in an event attracting an audience of over 200 featuring live EEG streaming during a violin performance by MacArthur Genius Fellow Johnny Gandelsman.

Interviewee/media article subject, UNC Health/School of Medicine **Dec 2024**
Interviewed for a feature article and video interview on 'Postdoc Journeys' at UNC Health/School of Medicine.

TV Interviewee, 10 News Australia **Dec 2023**
Participated in a TV news interview about a publication using advanced DTI (NODDI) to investigate structural connectivity during recovery from pediatric mild TBI.

Media article subject, *"Advanced imaging predicts the recovery of children with brain injury"* **Dec 2023**
Published in *UQ Queensland Brain Institute Media Release* and *The Hippocratic Post*

TV interview, 10 News Australia **Oct 2023**
I participated in a TV news interview about my personalised home tDCS headband and home-based clinical trial using tDCS to improve attention following acquired brain injury in children. This interview was conducted to promote entries into the Bionics Gamechangers Australia Student Innovation Challenge.